

# Amikosh Dube

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## INTRODUCTION

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Researching physical AI by integrating large language models, transformers, and vision systems to build general-purpose robots. Proven experience in simulation and perception, from founding a cloud-based drone simulation startup to engineering real-time vision detection pipelines in Unreal Engine.

## EDUCATION

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### Carnegie Mellon University

Master of Science in Computer and Civil Engineering

December 2026

Pittsburgh, PA

### Purdue University

Bachelor of Science in Computer Engineering

May 2025

West Lafayette, IN

## EXPERIENCE

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### AirLabs Researcher @ Carnegie Mellon University

August 2025 – Present

- Enhancing AirStack, an open-source drone simulation platform built in Isaac Sim
- Creating a drone solution to automatically monitor excavation progress for a Japanese construction firm, Shimizu

### Machine Learning Software Engineering Intern @ SAAB Defense

May 2023 – August 2025 (3 Summers)

- Engineering AI that uses federated learning and sensor data fusion to synthesize and convey situational information
- Trained a YOLO model using 10,000 synthetic images, achieving 89% precision against real-world drone datasets
- Engineered a real-time, asynchronous object detection pipeline in Unreal Engine using C++, via NNE and OpenCV
- Designed and implemented a custom topic-based messaging bus in Unreal Engine based on Kafka
- Devised an automated image pipeline that annotates objects to speed up the YOLO model training process
- Led cross-functional team meetings with university research collaborators to drive innovation

### Founder - Technical Lead @ Codex Labs LLC

May 2022 – Present

- Spearheaded the development of SWARMS, a cloud-based simulation platform for autonomous flight control, utilized by MIT Lincoln Laboratory, Alarm.com, and multiple researchers
- Designed an intuitive API and motion planning library that integrates with ROS
- Led customer outreach, product demonstrations, and strategic development to expand the platform's adoption

### Lead Drone SWARMS Researcher @ Purdue University

August 2021 – May 2025

- Built SLAM algorithms for autonomous drone navigation in complex environments
- Implemented real-time obstacle avoidance using LiDAR, Radars, and Cameras
- Lead the research team in collaboration with the professor to conduct meetings and assign tasks to students
- Presented at Purdue's Undergraduate Research Expo annually, advancing knowledge in robotic swarm control

## PROJECTS

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### Purdue Student Ticket Exchange

January 2025 – Present

- Developed a Discord server with 700+ members, providing a secure market for students to buy/sell student tickets

### VEX World Qualifier and Mentor

August 2017 – May 2025

- Four-time VEX Worlds Qualifier and top-50 ranked team; mentored high school teams to worlds placements

## SKILLS

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**Programming Languages** | C/C++, Python, SQL, Bash

**AI & Robotics** | Physical AI, Transformer Models, YOLO, SLAM, Sensor Fusion, PID/LQR

**Frameworks & Sim** | Docker, ROS2, Unreal Engine, NVIDIA Isaac Sim, AirSim

**Developer Tools** | Git, GitHub, VS Code, Visual Studio, Scrum, JIRA, Confluence, Source Control, UML Diagrams